

CREATE Module:
Perturbation Theory
Lecture 2: Regular Perturbations
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Example 3.1.1. (From Lin & Segel, page 186): Solve approximately

$$\begin{aligned}\epsilon x + 10y &= 21, \\ 5x + y &= 7,\end{aligned}\tag{3.1.1}$$

for $\epsilon = 0.01$.

- ✿ Here ϵ is a small parameter.
- ✿ Because ϵ is small the idea is to start with, hopefully, a simpler problem that results when $\epsilon = 0$, solve that and find corrections in terms of epsilon.

$$\epsilon = 0 \implies 10y_0 = 21 \implies y_0 = 2.1$$

$$y_0 = 2.1 \implies 5x_0 + 2.1 = 7 \implies x_0 = 0.98$$

$$\begin{aligned}x(\epsilon) &= x_0 + \epsilon x_1 + \epsilon^2 x_2 + \cdots, \\y(\epsilon) &= y_0 + \epsilon y_1 + \epsilon^2 y_2 + \cdots.\end{aligned}\tag{3.1.4}$$

Substituting these expansions into (3.1.16) gives

$$\begin{aligned}\epsilon^0(10y_0 - 21) + \epsilon(x_0 + 10y_1) + \epsilon^2(x_1 + 10y_2) + \cdots &= 0, \\ \epsilon^0(5x_0 + y_0) + \epsilon(5x_1 + y_1) + \epsilon^2(5x_2 + y_2) + \cdots &= 0.\end{aligned}\tag{3.1.5}$$

- ✿ The idea is to consider equations for various powers of epsilon as independent.
- ✿ The equations with 1 (or epsilon to the power of zero) as a coefficient are the equations we solved on the last slide.
- ✿ They are equivalent to setting epsilon=0 and are called the reduced problem or sometimes the leading order problem.
- ✿ The equations with epsilon to the power of 1 are called the first correction.

(a) The $\mathcal{O}(1)$ terms (those with coefficient ϵ^0) give

$$\begin{aligned} 10y_0 - 21 &= 0, \\ 5x_0 + y_0 &= 0. \end{aligned} \tag{3.1.6}$$

This is the reduced problem we have already solved.

(b) The $\mathcal{O}(\epsilon)$ terms give

$$\begin{aligned} x_0 + 10y_1 &= 0, \\ 5x_1 + y_1 &= 0. \end{aligned} \tag{3.1.7}$$

From this we find

$$y_1 = -\frac{x_0}{10} = -0.098, \tag{3.1.8}$$

and

$$x_1 = -\frac{y_1}{5} = 0.0196. \tag{3.1.9}$$

(c) The $\mathcal{O}(\epsilon^2)$ terms give

$$\begin{aligned} x_1 + 10y_2 &= 0, \\ 5x_2 + y_2 &= 0. \end{aligned} \tag{3.1.10}$$

giving

$$(x_2, y_2) = (0.000392, -0.00196). \tag{3.1.11}$$

Thus, to order ϵ^2 , we have

$$\begin{aligned} x &= 0.98 + 0.0196\epsilon + 0.000392\epsilon^2 + \dots, \\ x &= 2.1 - 0.098\epsilon - 0.00196\epsilon^2 + \dots. \end{aligned} \tag{3.1.12}$$

For $\epsilon = 0.01$ the first three terms give

$$(x, y) \approx (0.9801960392, 2.099019804). \tag{3.1.13}$$

✿ **The idea is that the equations at each order should be simpler to solve compared to the original problem (not really the case here).**

For $\epsilon = 0.01$ the first three terms give

$$(x, y) \approx (0.9801960392, 2.099019804). \quad (3.1.13)$$

The exact solution is

$$(x, y) = \left(\frac{49}{50 - \epsilon}, \frac{105 - 7\epsilon}{50 - \epsilon} \right), \quad (3.1.14)$$

which, for $\epsilon = 0.01$ gives

$$(x, y) = \left(\frac{49}{49.99}, \frac{104.93}{49.99} \right) = (0.9801960392 \dots, 2.09901980396 \dots). \quad (3.1.15)$$

- ✿ The perturbation solution was remarkably accurate!
- ✿ The reason that it works so well is that the solution depends on the parameter in a particular way.
- ✿ Ask yourself the question “what could go wrong in the exact solution as epsilon is varied?”
- ✿ A quick inspection shows that a singularity is present but only when epsilon=50.
- ✿ We were told at the outset that epsilon is a small parameter, so in this case there is nothing to worry about.
- ✿ When is there something to worry about?

Example 3.1.2. (From Lin & Segel): Find an approximate solution of the system

$$\begin{aligned}\epsilon x + y &= 0.1, \\ x + 101y &= 11,\end{aligned}\tag{3.1.16}$$

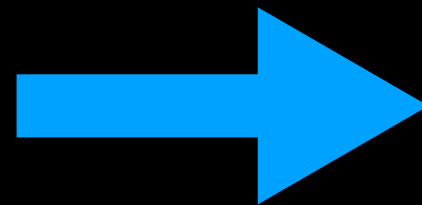
for $\epsilon = 0.01$.

- ✿ In the notes the perturbation solution is developed.
- ✿ Here I want to discuss just the exact solution, and how it shows that something goes wrong.

- ✿ So let's eliminate y to find:

$$101\epsilon x + 101y = 10.1$$

$$x + 101y = 11$$



$$(101\epsilon - 1)x = -0.9$$

$$x = \frac{-0.9}{101\epsilon - 1}$$

- ✿ The problem is that now a singularity results when $\epsilon = 1/101$!
- ✿ So a perturbation solution would work for $\epsilon=0.001$ but not for $\epsilon=0.01$ (too close to the singularity).

- ✿ Perturbation theory can be applied to many problems.
- ✿ Often (though not always) the first correction is the only thing of interest.
- ✿ The power series-like nature of perturbation theory leads to the terminology “regular perturbation theory” or RPT.
- ✿ Two classical fields of application useful to the applied mathematician are differential equations and polynomials.
- ✿ Polynomials are easier since the pure mathematical side of their solution is “settled” due to the Fundamental Theorem of Algebra (n solutions in the complex plane for an n th order polynomial).
- ✿ Hence we will treat these first (they also make the use of computer algebra, like Maple more transparent).

Problem 1

$$x^2 + (5 + \epsilon)x + 1 = 0$$

- Exact solution is easily found by quadratic formula.
 - Recall that the magnitude of epsilon is assumed to be smaller than one
 - Discriminant tells us there are two solutions: $D = (5 + \epsilon)^2 - 4$
- $$(x_0 + \epsilon x_1 + \dots)^2 + (5 + \epsilon)(x_0 + \epsilon x_1 + \dots) + 1 = 0$$
- $$(x_0^2 + 2\epsilon x_1 x_0) + 5x_0 + \epsilon(5x_1 + x_0) + 1 + \mathbf{h.o.t.} = 0$$
- Here the leading order term isn't that much simpler to solve since it doesn't factor.

$$x_0^2 + 5x_0 + 1 = 0 \implies x_0 = -\frac{5}{2} \pm \frac{\sqrt{21}}{2}$$

- But the first correction is easily determined in terms of the leading order solution:

$$2x_1x_0 + 5x_1 + x_0 = 0 \implies x_1 = -\frac{x_0}{2x_0 + 5}$$

Problem 2

$$x^2 = \epsilon \implies x = \pm \sqrt{\epsilon}$$

- OK let's see what happens for a (really simple) problem that is close to a double root.
- The reduced problem is in fact a double root problem, with $x=0$, being that double root.
- The exact solution shows that the approximate solution behaves as the square root of epsilon.
- This suggests our approximation should be written as: $x = x_0 + \sqrt{\epsilon}x_1 + \epsilon x_2 + \dots$

$$(x_0 + \sqrt{\epsilon}x_1 + \epsilon x_2 + \dots)^2 = \epsilon$$
$$x_0^2 + \sqrt{\epsilon}2x_0x_1 + \epsilon(2x_2x_0 + x_1^2) + \dots = \epsilon$$

- At leading order we recover the double root $x_0 = 0$, at the root epsilon order we get nothing because $x_0 = 0$, and finally at order epsilon we get:

$$x_1^2 = 1 \implies x = 0 \pm \sqrt{\epsilon}$$

- ✿ So far, so good. Let's doctor up a problem to see what could go wrong.
- ✿ We do this using a factored solution so we can read off the roots:

Problem 3 $(\epsilon x - 1)(x - 2) = 0 \implies x = 2, \frac{1}{\epsilon}$

- ✿ You can see that one root is 2 and has no dependence on epsilon while the other has a very strong dependence on epsilon.
- ✿ In fact when epsilon is small the root grows without bound!
- ✿ This is also evident in how the reduced problem behaves; it stops being a quadratic and becomes just $x=2$.
- ✿ So do we need to give up on RPT? May be not. Let's expand the polynomial and think about what we are trying to find.

$$\epsilon x^2 - (2\epsilon + 1)x + 2 = 0$$

$$\epsilon x^2 - (2\epsilon + 1)x + 2 = 0$$

- ✿ The leading order problem gave us the $x=2$ solution. It accomplished this by effectively “dropping” one of the terms in the equation.
- ✿ What we are missing is the “large” solution or the one that behaves like $1/\epsilon$
- ✿ So let’s see if we can make progress by choosing only some of the terms. This is called **“dominant balance”**.
- ✿ Physicists do this informally all the time by saying things like “neglecting the small effects of the Earth’s rotation”.
- ✿ The problem is that we have three possible choices or terms to drop.

$$\epsilon x^2 - (2\epsilon + 1)x + 2 \approx \epsilon x^2 - (2\epsilon + 1)x = 0$$

$$\epsilon \tilde{x} - (2\epsilon + 1) = 0 \implies \tilde{x} = \frac{1}{\epsilon} + 2 \approx \frac{1}{\epsilon}$$

- ✿ We need to check that the term we dropped is actually smaller than the two terms we kept: First term is $1/\epsilon$, second term is also like $1/\epsilon$ and the third term is a constant, so it is OK to drop as ϵ goes to zero.

$$\epsilon x^2 - (2\epsilon + 1)x + 2 = 0$$

- What if we did it “wrong”? Would we get a different answer, or is the process self-correcting in some way?
- Let's choose a balance between the first and third term: $\epsilon x^2 = -2$
- If epsilon is positive this gives an imaginary root: $x \approx i\sqrt{\frac{2}{\epsilon}}$
- A quick calculation shows that the first term is proportional to a constant, as is the third term.
- But the term we neglected is: $-2\sqrt{2}\sqrt{\epsilon}i - i\frac{\sqrt{2}}{\sqrt{\epsilon}}$
- As epsilon gets small the second term gets very large and this contradicts our assumption of this term being negligible.
- What about the second and third terms: $(2\epsilon + 1)x = 2$.
- This looks like an OK choice because: $x \approx \frac{1}{\epsilon}$
- But if that is the case the first term is $1/\epsilon$ and this grows without bound as epsilon goes to zero.
- Thus the balance we found on the last slide is our only choice.

Summary

- We have developed the regular perturbation theory for polynomials, using quadratics as a guide.
- Special techniques needed to be developed for repeated roots and the case when one of the roots was very large in the limit of the small parameter going to zero.
- For those interested I have examples of how Maple can be used to perform these expansions without the drudgery of doing the algebra by hand.
- There is one outstanding issue, and that is that we always just assumed the small parameter was there for us to use as we saw fit.

Say you want to solve the following equation using RPT

$$x^2 + 3.01x + 2.0002 = 0$$

- ✿ First we must introduce a small parameter
- ✿ It would be nice if we could write $\epsilon = 0.01$, the issue is what to do with 2.0002?
- ✿ Well $\epsilon^2 = 0.0001$ so we have $3.01 = 3 + \epsilon$ and $2.0002 = 2 + 2\epsilon^2$

$$x^2 + (3 + \epsilon)x + 2 + 2\epsilon^2 = 0$$

- ✿ An now we can just apply RPT:

$$(x_0 + \epsilon x_1 + \epsilon^2 x_2)^2 + (3 + \epsilon)(x_0 + \epsilon x_1 + \epsilon^2 x_2) + 2 + 2\epsilon^2 = 0$$

- ✿ Leading order: $x_0^2 + 3x_0 + 2 = 0 \implies x_0 = -2, -1$

- ✿ First order: $2x_0x_1 + 3x_1 + x_0 = 0 \implies x_1 = -\frac{x_0}{2x_0 + 3}$

- ✿ Second order: $2x_2x_0 + x_1^2 + 3x_2 + x_1 + 2 = 0 \implies x_2 = -\frac{x_1^2 - x_1 - 2}{2x_0 + 3}$

RPT for Simple ODEs

The good and the bad

Problem 1

Consider the simple DE for the decaying exponential

$$\frac{dy}{dx} = -y$$

With the perturbed initial condition $y(0) = 1 + 2\epsilon$

Try RPT so that $y(x) = y^{(0)} + \epsilon y^{(1)} + O(\epsilon^2)$

Substitute into the DE to get

$$\frac{dy^{(0)}}{dx} + \epsilon \frac{dy^{(1)}}{dx} + O(\epsilon^2) = -y^{(0)} - \epsilon y^{(1)} + O(\epsilon^2)$$

And the ICs expand as $y^{(0)}(0) + \epsilon y^{(1)} + O(\epsilon^2) = 1 + \epsilon \cdot 2 + O(\epsilon^2)$

At leading order we get

$$\frac{dy^{(0)}}{dx} = -y^{(0)} \text{ with } y^{(0)}(0) = 1 \text{ and this has the solution } y^{(0)} = \exp(-x).$$

For the first order problem we get:

$$\frac{dy^{(1)}}{dx} = -y^{(1)} \text{ with } y^{(1)}(0) = 2 \text{ which can be solved to give } y^{(1)} = 2 \exp(-x).$$

The approximate solution thus reads

$$y(x) = \exp(-x) + \epsilon 2 \exp(-x) + O(\epsilon^2)$$

And a quick calculation confirms that we have recovered the exact solution:

$$y_{\text{exact}}(x) = (1 + 2\epsilon)\exp(-x)$$

Problem 2

Consider the DE for the decaying exponential with a small non zero “forcing”

$$\frac{dy}{dx} = -y + \epsilon(\sin(x) - \cos(x)) \text{ with the initial condition } y(0) = 1.$$

Try RPT so that $y(x) = y^{(0)} + \epsilon y^{(1)} + O(\epsilon^2)$

Substitute into the DE to get

$$\frac{dy^{(0)}}{dx} + \epsilon \frac{dy^{(1)}}{dx} + O(\epsilon^2) = -y^{(0)} - \epsilon y^{(1)} + \epsilon(\sin(x) - \cos(x)) + O(\epsilon^2)$$

And the ICs expand as $y^{(0)}(0) + \epsilon y^{(1)} + O(\epsilon^2) = 1 + \epsilon \cdot 0 + O(\epsilon^2)$.

At leading order we get $\frac{dy^{(0)}}{dx} = -y^{(0)}$ with $y^{(0)}(0) = 1$, and this has the solution

$$y^{(0)} = \exp(-x).$$

For the first order problem we get:

$$\frac{dy^{(1)}}{dx} = -y^{(1)} + \sin(x) - \cos(x) \text{ with } y^{(1)}(0) = 0.$$

The solution to this problem can be written as $y^{(1)} = y_{GS} + y_{PI}$ or the sum of a general solution of the homogeneous problem and a particular integral. Solving gives

$$y^{(1)}(x) = c_1 \exp(-x) - \cos(x) \text{ and applying the initial condition gives } c_1 = 1.$$

The approximate solution thus reads $y(x) = \exp(-x) + \epsilon(\exp(-x) - \cos(x))$

The solution can be written in a slightly more revealing way:

$$y(x) = (1 + \epsilon)\exp(-x) - \epsilon \cos(x)$$

You see that the perturbation leads to a small change in the decaying exponential and for long times, to a small oscillating part of the solution.

For a problem where an exact solution is not available, this sort of qualitative conclusion would be a worthy goal for RPT

Problem 3

Consider the simple DE for a perturbed decaying exponential

$$\frac{dy}{dx} = -(1 + \epsilon)y \text{ with the initial condition } y(0) = 1.$$

Try RPT so that $y(x) = y^{(0)} + \epsilon y^{(1)} + O(\epsilon^2)$. Substitute into the DE to get

$$\frac{dy^{(0)}}{dx} + \epsilon \frac{dy^{(1)}}{dx} + O(\epsilon^2) = -y^{(0)} - \epsilon y^{(1)} - \epsilon y^{(0)} + O(\epsilon^2)$$

And the ICs expand as $y^{(0)}(0) + \epsilon y^{(1)} + O(\epsilon^2) = 1 + \epsilon \cdot 0 + O(\epsilon^2)$

At leading order we get $\frac{dy^{(0)}}{dx} = -y^{(0)}$ with $y^{(0)}(0) = 1$.

This has the solution $y^{(0)} = \exp(-x)$.

For the first order problem we get:

$$\frac{dy^{(1)}}{dx} = -y^{(1)} - y^{(0)} = -y^{(1)} - \exp(-x) \text{ with } y^{(1)}(0) = 0$$

Again write $y^{(1)} = y_{GS} + y_{PI}$ and solve to get $y^{(1)}(x) = c_1 \exp(-x) - x \exp(-x)$ and applying the initial condition gives $c_1 = 0$.

The approximate solution thus reads $y(x) = \exp(-x) + \epsilon(-x \exp(-x))$.

Is this an OK approximate solution? Consider the absolute value of the correction term divided by the leading order term: $|-x \exp(-x)| / |\exp(-x)| = x$ which increases without bound as x gets larger and larger.

This means that RPT has failed, because the correction must be smaller than the leading order solution. Let's write out the exact solution to see why things may have gone wrong:

$$y_{exact} = \exp(-[1 + \epsilon]x) = \exp(-x)\exp(-\epsilon x)$$

The second term decays much more slowly than the first and hence we suspect RPT could fail if MORE THAN ONE scale appears in the problem.